

LAB 2

Website Layout & Frontend Foundation

Lab Title

Frontend Structure and Responsive Layout for the Semester Project

1. Lab Overview

In this lab, students will design and implement the **frontend structure of their semester project website**.

The website will represent a **generic online item catalog**, where items could represent books, clothing, electronics, food, or courses depending on each student's assigned domain.

The focus of this lab is **frontend development**, including:

- HTML5 semantic layout
- CSS3 styling
- Bootstrap responsive design
- basic JavaScript interaction

This lab corresponds to the **design phase of the Software Development Life Cycle (SDLC)**.

2. Learning Objectives

After completing this lab, students will be able to:

- design a structured website layout using HTML5
- build responsive pages using Bootstrap
- organize project folders properly
- create reusable UI components
- implement basic JavaScript interaction
- prepare the frontend structure required for the semester project

3. Prerequisites

Students should have the following tools:

- XAMPP or MAMP
- Code Editor
- Browser

and basic knowledge of:

- HTML
- CSS
- JavaScript

4. Lab Tasks

Task 1 — Create the Project Folder Structure

- Create a **project folder** for your semester project: **project_name**

Inside the folder create the following structure:

```
project_name
├── public
│   ├── index.php
│   ├── css
│   ├── js
│   └── images
├── pages
└── components
```

Explanation:

Folder	Purpose
public	main web pages
css	stylesheets
js	JavaScript files
images	images used in the website
pages	additional pages
components	reusable components (header, footer, etc.)

Task 2 — Create the Homepage Layout

Create file: **public/index.php**

Add a basic HTML5 structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Item Catalog Website</title>
  </head>
  <body>
    <h1>Welcome to the Website</h1>
  </body>
</html>
```

Task 3 — Implement Header and Navigation Bar

Add a header section containing:

- website **logo**
- navigation menu

Example navigation items:

- Home
- Items
- Search
- Contact

Example:

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand">My Website</a>
    <ul class="navbar-nav">
      <li class="nav-item"><a class="nav-link">Home</a></li>
      <li class="nav-item"><a class="nav-link">Items</a></li>
      <li class="nav-item"><a class="nav-link">Search</a></li>
      <li class="nav-item"><a class="nav-link">Contact</a></li>
    </ul>
  </div>
</nav>
```

Task 4 — Add Bootstrap

Add Bootstrap CDN to the <head> section:

```
<link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/boot  
strap.min.css" rel="stylesheet">
```

Task 5 — Create Item Catalog Layout

Create a section displaying items using Bootstrap cards.

Example structure:

```
<div class="row">  
  <div class="col-md-4">  
    <div class="card">  
        
      <div class="card-body">  
        <h5>Item Name</h5>  
        <p>Short description</p>  
        <button class="btn btn-primary">View  
                                Details</button>  
      </div>  
    </div>  
  </div>  
</div>
```

Create **at least six item cards**.

Task 6 — Create Item Details Page

Create file: [pages/item_details.php](#)

Display information such as:

- item name
- image
- description
- price
- category

Example:

```
<h2>Item Name</h2>
<p>Price: $100</p>
<p>Item description</p>
```

Task 7 — Add Basic JavaScript Interaction

Create file: `public/js/script.js`

Example interaction:

```
function showMessage() {
  alert("Item added");
}
```

Attach to a button.

5. Expected Results

Students should have:

- homepage with header and navigation bar
- responsive layout using Bootstrap
- item catalog page
- item details page
- basic JavaScript interaction

6. Deliverables

Students must demonstrate:

- homepage layout
- navigation bar
- item catalog UI
- responsive design